SAI AAKARSH PADMA

+1 (979) 739-8560 | saiaakarshp@gmail.com | College Station, TX, USA | LinkedIn | GitHub

EDUCATION

Texas A&M University - College Station **Aug 2024 - May 2026**
Master's, Computer Science
- Software Engineering, Machine Learning, Artificial Intelligence, Algorithms, NLP, Operating Systems, Network Security

Indian Institute of Technology Kharagpur (IITKGP) **Jul 2018 - May 2022**
Bachelor's, Electronics and Communication Engineering *GPA: 8.48*

SKILLS

Languages: Python, Java, C/C++, JavaScript, Ruby on Rails, HTML/CSS
Libraries/Frameworks: Spring Boot, React.js, Docker, Node.js, Flask, Lambda, IVS, GraphQL, AngularJS, Django
Databases: MongoDB, Redis, SQL, Kafka, Memcached, Postgres, MySQL, Oracle, Scylladb
Concepts: Data Structures & Algorithms, Agile, SCRUM, Web Development, Kubernetes, REST APIs, Image Processing
Tools: JIRA, Confluence, Excel, Git, Jenkins, Apache Spark, AWS, Azure, Google CLoud Platform

WORK EXPERIENCE

Exo Imaging **Santa Clara, CA, USA**
Imaging Systems Intern *Jul 2025 - Present*
- Optimized tracking algorithms with SSD & OpenCV to cut processing time from 30s to 0.2s, while improving usability with configurable label displays, multi-image visualization, tab-based navigation, and automated PDF reporting.
- Strengthened deployment with encrypted JSON handling and configurable SDK packaging, and integrated advanced analytical capabilities using anatomical indices to expand functionality and support flexible production use.

Texas A&M University **College Station, TX, USA**
Full Stack Developer *Apr 2025 - Present*
- Developed and maintained Dr. Bruce Wang's wisdomlinked.com chat application using React, Node.js, and MongoDB, adding features like user management, payment processing, guest access, file drop-off, and shareable join links.
- Enhanced platform functionality by enabling third-person participation in single calls, integrating feedback systems and email notifications, and troubleshooting critical bugs to ensure reliable performance and a seamless user experience.

ZEE **Bangalore, KA, India**
SDE1 *Jul 2022 - Mar 2024*
- Developed and launched "Houzee," a game service from concept to production, managing the full lifecycle while integrating Server-Sent Events for real-time updates and Kafka pipelines, ensuring smooth user experience and scalable performance.
- Engineered Java backend services with Spring Boot, optimizing throughput from 25tps to 200tps, and improved infrastructure by replacing Redis with Memcached, reducing API response times and simplifying configurations.
- Led the migration from AWS to GCP by debugging critical services, resolving complex integration issues, and coordinating cross-service transitions, ensuring uninterrupted operations and a 100% successful platform switch with long-term scalability.

PROJECTS

Agent-Based COVID-19 Modeller for University Campuses
- Developed an agent-based model simulating COVID-19 spread on a 2,200-student campus, using timetable and survey data for contact graphs and implementing intervention strategies that reduced infection rates by up to 30%.
- Built and deployed an LSTM-based emulator to accelerate simulations, analyzing performance with time-series graphs that highlighted strong initial alignment with the full simulator, followed by gradual information loss over time.

SAFAR
- Developed Flask APIs and a React front end to integrate YOLO-based object detection with video uploads, enabling traffic data analysis through interactive graphs, logs, and a user-friendly interface for seamless visualization.

Hybrid Self-RAG: Factual Consistency in Large Language Models
- Designed an LLM pipeline with retrieval-augmented generation, self-reflective decoding, and external validation, and built FactScore using SentenceTransformer embeddings and FAISS to evaluate claim–evidence alignment.
- Enhanced factual reliability by applying dynamic reranking and regeneration strategies with GPT-4, Hugging Face, and spaCy on NVIDIA A100 GPUs, improving factual accuracy to 89% while reducing hallucination rates by 42%.

ELRC
- Engineered an admin dashboard for visualizing and managing survey data across multiple schools, integrating role-based access control and secure pipelines to ensure accurate reporting and safe handling of sensitive information.
- Built a system that compares survey results and generates a dynamic tetrahedron model to visualize distortions.